Technology Validation and Start Up Fund

Round 32 Proposal Evaluations

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1) Executive Summary

Redwood is a Columbus, Ohio based LLC founded by former Battelle executives over 10 years ago. Redwood has assembled an extraordinary team for this Program. Each member of the five-person Redwood team is an accomplished technology commercialization professional with decades of experience in performing business and technical evaluations. This team, combined with identified external subject matter experts, has extensive experience in all six of the Ohio Third Frontier technology focus areas. More detail on the Redwood team is provided in Appendix 1 of this report and on our website (www.Redwdinnnov.com). Details of the TVSF program and the review process are provided in Appendix 2.

Eleven (11) TVSF Round 32 applications, totaling $1,350,000, were received and evaluated.

The proposals consisted of 1 Phase 1 application totaling $100,000 (1 Phase 1, Option 3 applicant of $100,000) and 10 Phase 2 applications totaling $1,250,000 for a combined total of $1,350,000.

Funding is recommended for the one Phase 1 application totaling $100,000 and six Phase 2 applications for a total of $750,000 yielding a combined total of $850,000. This translates to a 64% recommended application funding rate for this TVSF round, compared to the average of 49% over all 32 TVSF rounds. The recommended approval rate for Phase 2 proposals in this round is 60%.

In addition, the Phase 1 Round 30 application from Nationwide Children's Hospital that had previously been recommended for funding of $100,000 has been resubmitted with revisions. This resubmitted application included a request for an additional $100,000 funding - total requested funding of $200,000. *This resubmitted application is not included in the metrics for Round 32. It is described in the one-page document on Page 18.*

2) Evaluation Results

Summaries of the evaluations of the proposals for Round 32, incorporating the interview, and funding recommendations are shown below in Tables 1 and 2. Rounds 22-29 interviews were conducted via video due to COVID-19 concerns. Round 30 interviews were conducted in person and Rounds 31 and 32 were conducted via video. The total recommended funding for Phase 1 and Phase 2 projects is, respectively, $100,000, and $750,000. Note that the Table 1 and 2 column widths are proportional to the weighting of the evaluation criteria. For example, in Table 1, Selection Committee which is weighted at 20 is four times as wide as project selection which is weighted at 5. Note that a yellow evaluation indicates that the proposal meets that particular criterion.

More detailed evaluations and recommendations for each Phase 1 and Phase 2 proposal may be found in Section 3 of this report.

Table 1 – Phase 1 Proposal Evaluation and Funding Recommendation

Table 1
Phase 1 Proposal Evaluation and Funding Recommendation
TVSF Round 32

Proposal #	Applicant	Requested Funding ($1,000)	Recommended Funding ($1,000)	Selection Committee	Deal Flow & Budget Strategy	External Participation	Track Record	Metrics	Project Management & Experience	Project Selection Process
FY23-4226	University of Akron Research Foundation	100	100							
	Total	100	100	Column width is proportional to score weighting in each category						
Evaluation Scale				Weak		Meets		Exceeds		

Option 1: $200,000 to $500,000, 1:1 cost share, 1 year to allocate funds, 2 year project period.

Option 2: $1,000,000, 1:1 cost share, 2 years to allocate funds, 3 year project period.

Option 3: $100,000, internship component, 1 year to allocate funds, 2 year project period.

Table 2 – Phase 2 Proposal Evaluation and Funding Recommendation

Table 2
Phase 2 Proposal Evaluation and Funding Recommendation
TVSF Round 32

Proposal Number	Lead Applicant	Requested Funding ($1,000)	Recommended Funding ($1,000)	Team	Opportunity/Market Size	IP Protection	Proof of Concept	Potential Investor & Business Partner Engagement	Business Model	Project Plan/Budget Narrative	Growth Plan in Ohio	ESP Interaction
FY23-4229	Enspire DBS Therapy, Inc.	150	150	Exceeds	Exceeds	Meets	Exceeds	Exceeds	Exceeds	Meets	Meets	Meets
FY23-4230	HydroSmart, LLC	100	100	Meets	Exceeds	Meets	Exceeds	Exceeds	Meets	Exceeds	Exceeds	Exceeds
FY23-4231	Mobius Care, Inc.	150	150	Exceeds	Exceeds	Meets	Exceeds	Exceeds	Meets	Exceeds	Meets	Meets
FY23-4232	Neuright, Inc.	150	150	Meets	Meets	Meets	Exceeds	Exceeds	Meets	Exceeds	Meets	Exceeds
FY23-4235	S&J NanoChemicals, Inc	100	100	Meets	Meets	Meets	Meets	Meets	Meets	Meets	Meets	Meets
FY23-4236	Zsense Systems	100	100	Exceeds	Meets	Meets	Meets	Meets	Exceeds	Meets	Meets	Meets
	Sub-Total	750										
FY23-4227	Ace Medical Company	150	-	Weak	Weak	Weak	Weak	Weak	Weak	Weak	Meets	Weak
FY23-4228	Advanced and Innovative Multifunctional Materials, LLC	150	-	Meets	Weak	Meets	Meets	Meets	Meets	Weak	Meets	Meets
FY23-4233	Phantom Solutions	100	-	Weak	Weak	Meets	Weak	Weak	Meets	Weak	Meets	Meets
FY23-4234	Raider Technologies, LLC	100	-	Weak	Exceeds	Weak	Meets	Meets	Weak	Meets	Exceeds	Exceeds
	Sub-Total	500		Column width is proportional to score weighting in each category								
	Total	1,250										

Evaluation Scale	Weak	Meets	Exceeds

Table 3 lists the funding approval rate by TVSF round. This round's approval rate of 64% of the total submitted proposals is on the high side of the average over all thirty-two rounds. The Phase 1 application recommended for approval, comprised 9% of the submittals for this round. The historical range of recommendations for individual rounds has spanned 27 – 100%, with an average of 49%.

Table 3. TVSF Approval Rate by Round

TVSF Round 32
Approval Rate by Round
Phase 1/Phase 2

Round	$ Recommended	Approval Rate
1 (APR 2012)	$950,000	35%
2 (AUG 2012)	$900,000	52%
3 (DEC 2012)	$610,000	44%
4 (JUN 2013)	$864,000	30%
5 (FEB 2014)	$1,462,000	46%
6 (JUN 2014)	$998,000	39%
7 (OCT 2014)	$1,100,000	57%
8 (FEB 2015)	$710,000	37%
9 (JUN 2015)	$550,000	31%
10 (DEC 2015)	$925,000	38%
11 (APR 2016)	$1,239,000	46%
12 (OCT 2016)	$3,537,269	46%
13 (MAR2017)	$1,567,500	38%
14 (SEP 2017)	$498,832	27%
15 (DEC 2017)	$2,250,000	38%
16 (MAR 2018)	$2,098,600	52%
17 (SEP 2018)	$2,100,000	42%
18 (DEC 2018)	$1,150,000	35%
19 (APR 2019)	$2,250,000	43%

Round	$ Recommended	Approval Rate
20 (NOV 2019)	$1,350,000	43%
21 (FEB 2020)	$3,944,000	56%
22 (JUN 2020)	$1,398,630	53%
23 (DEC 2020)	$900,000	50%
24 (MAR 2021)	$2,092,900	55%
25 (JUN 2021)	$800,000	75%
26 (OCT 2021)	$1,700,000	55%
27 (FEB 2022)	$850,000	43%
28 (APR 2022)	$2,499,976	64%
29 (JULY 2022)	$850,000	100%
30 (OCT 2022)	$3,700,000	71%
31 (JAN 2023)	$100,000	50%
32 (APR 2023)	$850,000	64%
Total Funding	**$46,795,707**	
Average/Round	**$1,462,366**	**49%**

3) Proposal Summaries

Phase 1 Proposal Summaries and Recommendations

Proposal 23-4226	**University of Akron Research Foundation**	Amount Requested: $100,000
Prior Phase 1 Applications: Yes	*University of Akron Research Foundation Spark Fund*	Amount Recommended: $100,000

TOTAL BUDGET: $100,000

Institution Snapshot: The University of Akron Research Foundation (UARF) Spark Fund will provide the proof needed to bring University of Akron (UA) technologies in areas of institutional strength to the point where a scalable startup can be built based on the technology.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
G	**Selection Committee**	Well-rounded 7-member committee with experience in managing startups; entrepreneurial, investor backgrounds and knowledge in some of the technical areas targeted by Spark Fund.
Y	**Deal Flow; Budget Strategy**	Deal flow of 15 projects is consistent to fund 3 to 4 projects for amount of funding requested and have an impact on the project outcome.
Y	**External Participation**	Funds are controlled by UARF and distributed directly to third party providers. ESP, local polymer organizations and NEOMED among others are resources upon which Spark Fund rely.
G	**Track Record**	UARF received a Phase 1 grant in 2022. Four projects were funded under that Phase 1 grant.
Y	**Metrics**	Assumes 4 projects lead to 2 licenses to start-ups. Seems reasonable based on track record. Encourage UARF team to develop quantitative financial goals.
G	**Project Management & Experience**	Projects are managed by UARF personnel associated with Spark Fund who have management experience.
G	**Project Selection**	A clear, well-developed process for selecting projects is used.
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds

Comments and Recommendation: UARF has a clear well-developed process for selecting projects and for obtaining results with limited resources. There have been a few changes to the selection committee. If further changes in selection committee are made, UARF is encouraged to choose members more closely aligned with Akron's strength and focus areas. The applicant is recommended for funding.

Proposal Summaries - Phase 2 Recommended for Funding

Proposal 23-4229	**Enspire DBS Therapy, Inc**	Amount Requested: $150,000
Licensing Institution	Cleveland Clinic Foundation	Amount Recommended: $150,000
Prior Phase 1 Applications: No	Prior Phase 2 Applications: No	***DBS for Stroke Program Optimization System***

Company Snapshot: Enspire's therapy is focused on optimizing the delivery of Deep Brain Stimulation (DBS) to the Dentate Nucleus (DN) of the cerebellum in order to improve arm function in patients with chronic deficits due to stroke.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
G	Management Team	The team qualifications are strong from the market, therapy and entrepreneurial perspectives. Further, team size and time commitments are appropriate.
G	Opportunity/Market Size	Total US market for just the optimizer is ~ $300M. The market for the full DBS system is many times larger.
Y	Intellectual Property Protection	There is a 2021 patent application that is specific to the optimization device and approach.
G	Proof of Concept	Proof of concept is appropriate and well described in supplementary materials.
G	Potential Investor/ Business Partner Engagement	Extensive strategic investor engagement has occurred and an investment is imminent.
G	Business Model	The therapy creates substantial economic value and payment codes exist. A reasonable strategy to bridge clinical trial to launch is given.
Y	Project Plan/ Budget Narrative	Project plan is appropriate with a credible budget.
Y	Growth Plan in Ohio	Company is committed to grow in Ohio with 5 -10 max. employees.
Y	ESP Interaction	Company is engaged with JumpStart.
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds

Comments and Recommendations: This application does an excellent job of identifying how a specific TVSF investment will truly enable a much more expensive medical device development. In addition, the team is strong and the value creation potential – both economic and human health is compelling. This application is recommended for funding.

Proposal 23-4230	**HydroSmart, LLC**	Amount Requested: $100,000
Licensing Institution	US Army	Amount Recommended: $100,000
Prior Phase 1 Applications: No	Prior Phase 2 Applications: No	***Innovative Fluid Consumption Measurement and Reporting Solution (HydroSmart)***

Company Snapshot: HydroSmart, LLC plans to develop accurate, effective, automated, and user-friendly measurement of fluid intake via a wearable sensor system.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
Y	Management Team	Strong contract team assembled to execute the proposed project. Additional commercial and technical staff will be needed to achieve the project growth plan for the company.
G	Opportunity/Market Size	Large market opportunity. TAM "USA Consumer Wellness" target market > $1.0 billion. TAM "USA Military" target market > $0.5 billion. Proposal notes potential near term supply contract with Army.
Y	Intellectual Property Protection	Negotiating license under US Army Patent 11,293,795 for exclusive field of use for "Consumer Wellness" market. Proposal states that there is significant trade secret IP in design of the device.
G	Proof of Concept	Device tested by US Army @ TRL 6. Proposal states "company has developed detailed bill of materials and manufacturing costs" for the prototype device.
G	Potential Investor/ Business Partner Engagement	Potential SBIR/CRADA funding. Several potential funding groups have been approached by management team.
Y	Business Model	Sales revenue from hardware and software license. Projected Sales Revenue $52 MM, GM = 88%, NM = 40% (Year 5).
G	Project Plan/ Budget Narrative	Contractors have offered fixed price contracts to build prototype.
G	Growth Plan in Ohio	Project > 200 employees with > $10 million payroll in five years.
G	ESP Interaction	Working closely with the Entrepreneurs' Center in Dayton.
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds

Comments and Recommendations: Project team includes equipment design and software contractors with strong technical expertise. These contractors will work under fixed price contracts, derisking the project. Without participation of these contractors, project risk is considered significantly higher. Additional commercial and technical staff will be needed to grow company. Initial target market "USA Consumer Wellness" has estimated TAM > $1.0 billion. Follow-up target market "USA Military" has estimated TAM > $0.5 billion. Proposal reports strong market pull for military market with potential funding under CRADA/SBIR grants.

Proposal 23-4231	**Mobius Care, Inc.**	Amount Requested: $150,000
Licensing Institution	Cleveland Clinic Foundation	Amount Recommended: $150,000
Prior Phase 1 Applications: No	Prior Phase 2 Applications: No	***Small Molecule IBD Therapeutics***

Company Snapshot: Mobius Care, Inc. was formed to develop treatments for inflammatory bowel disease (IBD), targeting three intertwined dysfunctions in patients: dysfunctional inflammatory system, dysfunctional wound healing and dysfunctional microbiome.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
G	**Management Team**	An outstanding team of accomplished and well-connected medical scientists, physicians, managers and investors.
G	**Opportunity/Market Size**	The global IBD market valued at $21 billion in 2021, projected to reach $34.5 billion by 2031. Conservative estimate of the serviceable market in the US is $2-3 billion.
Y	**Intellectual Property Protection**	Six patent applications are cited, two for each area of Mobius' product development pipeline.
G	**Proof of Concept**	The milestones are clearly described and (likely) achievable in the timeframe. Dollar estimates are quotes from vendors.
G	**Potential Investor/ Business Partner Engagement**	Initial discussions demonstrated a strong interest from the venture community. Core team are, themselves, serial entrepreneurs.
Y	**Business Model**	High risk - high reward drug repurposing an approved, off-patent drug for a new indication via the FDA 505(b)2 regulatory pathway.
G	**Project Plan/ Budget Narrative**	Development of gut restricted orally available reformulation.
Y	**Growth Plan in Ohio**	Ohio-based startup.
Y	**ESP Interaction**	The Mobius team has engaged with JumpStart.
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	**Weak**	**Meets**	**Exceeds**

Comments and Recommendations: Mobius Care team is comprised of committed and highly experienced scientist-entrepreneurs. The focus of the TVSF application is reformulation of an approved antifungal drug to treat a newly discovered source of persistent fungal infection in the intestines of Inflammatory Bowel Disease patients. Reformulation and repurposing of an approved, off-patent drug could reduce time and cost to reach clinical trials. This application is recommended for funding.

Proposal 23-4232	**Neuright, Inc.**	Amount Requested: $150,000
Licensing Institution	University of Maine	Amount Recommended: $150,000
Prior Phase 1 Applications: No	Prior Phase 2 Applications: 31	***First in human validation of a novel medical diagnostic device for peripheral neuropathy, to support commercialization***

Company Snapshot: Neuright is developing a device to diagnose peripheral neuropathy (PN). Subsequent versions of the device will be developed to treat PN.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
Y	**Management Team**	Highly qualified technical team. Consultants and support from Rev1. Outside service providers and experts round the team sufficiently in medical device development and corporate management.
Y	**Opportunity/Market Size**	Global treatment market ~$1.8B today, CAGR ~3.6%. An estimated 30 million people in the U.S. have PN.
Y	**Intellectual Property Protection**	"Methods and Devices for Treatment of Neuropathy" (PCT/US21/31337) nationalized in several countries. In negotiations with University of Maine to obtain an exclusive license agreement.
G	**Proof of Concept**	Unmet need is clearly described, which available technology does not address adequately. Device quantities sufficient for clinical studies will be manufactured. Outputs are clearly defined.
G	**Potential Investor/ Business Partner Engagement**	Interest from Rev1 and Maine Angels. Successful completion of TVSF project will provide data to seek follow-on funding.
Y	**Business Model**	Target customers defined. Path to approval focused on customers and needs of investment community.
G	**Project Plan/ Budget Narrative**	The project plan is clearly described and achievable in 12 months.
Y	**Growth Plan in Ohio**	Officers, manufacturing and future development in Central Ohio.
G	**ESP Interaction**	Engaged with Rev1, who provide frequent advice and support.
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds

Comments and Recommendations: The founders have captured the clinical unmet needs and described how their technology will address those needs. Budget and aims are well organized and achievable. Business support and business model refinement much improved as a result of participation from Rev1, consultants and outside service providers. This collaboration should continue. Neuright's current and future business plans demonstrate their commitment to maintaining their principal place of business in Central Ohio. This application is recommended for funding.

Proposal 23-4235	**S&J NanoChemicals, Inc.**	Amount Requested: $100,000
Licensing Institution	University of Dayton	Amount Recommended: $ 100.000
Prior Phase 1 Applications: No	Prior Phase 2 Applications: 30	***ZincX sustainable nanofertilizers for enhanced crop growth***

Company Snapshot: The company has developed two novel nanofertilizers that enhance growth, yield, and vitality in a wide variety of crops in agricultural production through a unique and environmentally friendly application strategy.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
Y	**Management Team**	Strong technically in the field. External consultant has business development, fund raising, and marketing skills
Y	**Opportunity/Market Size**	US market for micronutrient based nano fertilizers is estimated to be at $ 148 million/year with good growth prospects for the US.
Y	**Intellectual Property Protection**	Filed for a patent application in Nov 2022.
Y	**Proof of Concept**	Results show marked increase in growth rate, yield for corn/ peas.
Y	**Potential Investor/ Business Partner Engagement**	Awaiting TVSF funded field trial results to initiate fund raising. Have raised $ 40 K in Angel funding.
Y	**Business Model**	3-prong business model- targeting farmers for crops, ag companies, selling coated seeds and licensing. investments in years 2/3
Y	**Project Plan/ Budget Narrative**	Three major milestones and tasks will be project targets.
Y	**Growth Plan in Ohio**	Plan to grow the business in Dayton area; 3 FT, 3 PT staff in yrs 2/3.
Y	**ESP Interaction**	Working with local ESP centers and staff.
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds

Comments and Recommendations: S&J NanoChamicals, Inc. has a very strong technical Team with proven experience in the nano fertilizer field. Addition of an external advisor- DSLV ventures LLC at 1 day/ week effort- seems to provide needed business development and fund raising skills. Market and opportunities for targeted nano fertilizer for crops with improved growth rate, yield and reduced environmental foot print are real and growing. A successful deployment of the early stage technology should have a major impact on crops such as soybeans and corn—a significant part of Ohio farmers crops. The Team is encouraged to focus their efforts on working with local markets/ end users and not dilute their limited sources to overseas targets.

Proposal 23-4236	**ZSense Systems, LLC**	Amount Requested: $100,000
Licensing Institution	University of Akron Research Foundation	Amount Recommended: $100,000
Prior Phase 1 Applications: No	Prior Phase 2 Applications: No	***Inline Oil Condition Monitoring System***

Company Snapshot: ZSense Systems, LLC is developing an oil condition monitoring system to 1) monitor the early onset wear progression in real time with higher resolution than competitors and 2) predict remaining useful life of every at-line machine by accurately detecting multiple important oil properties.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
G	Management Team	A strong and well-balanced team with a business mentor and industry consultant providing support to the team.
Y	Opportunity/Market Size	Serviceable market size is estimated at $ 160 MM/ yr. with CAGR of 9.5%. Initial market focus is $ 107 MM/ yr. in 2023.
Y	Intellectual Property Protection	U.S Patent 8,522,604 to be licensed from Akron Research Foundation. An earlier patent US Patent 7,777,476 is part of portfolio.
Y	Proof of Concept	Current TRL is 4.
Y	Potential Investor/ Business Partner Engagement	None at this time; however, have needed contacts and interactions to engage early-stage adaptors/ investors.
G	Business Model	Well laid out business model address the target mid-size market with product sales and yearly service fee. Revenue to start in year 4/5.
Y	Project Plan/ Budget Narrative	Seems reasonable to move to TRL 5.
Y	Growth Plan in Ohio	Plan to grow business in Dayton area with 5-10 staff in year 5.
Y	ESP Interaction	Resident of Bounce Hub; active with UARF for guidance/ mentoring.
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds

Comments and Recommendations: A well balanced Team with technical, engineering, marketing and business development skills. Has demonstrated ability to secure non diluted funds. Patent appears to be strong. The business model is well laid out with market segmentation and 3 revenue paths. The Team is strongly urged to stay focus on rapidly demonstrating their integrated sensor system at file sites and move on to commercialization.

Proposal Summaries - Phase 2 Not Recommended for Funding

Proposal 23-4227	**ACE Medical Company**	Amount Requested: $150,000
Licensing Institution	Cleveland Clinic Foundation	Amount Recommended: $0
Prior Phase 1 Applications: No	Prior Phase 2 Applications: No	***Bone Duster***

Company Snapshot: ACE Medical is developing a device that collects bone fragments to be used in lumbar spine surgery.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
R	**Management Team**	A generic description of ACE Medical employees and specializations was provided in the proposal.
R	**Opportunity/Market Size**	Market segment of Global Spinal Implants and Surgery Devices was valued, not Bone Grafts and substitutes. No Total Addressable market identified or valued.
R	**Intellectual Property Protection**	IP stated as not yet filed in proposal.
R	**Proof of Concept**	No description, diagram provided in proposal. Currently made from components in the hospital by a surgical technician, who is also the inventor.
R	**Potential Investor/ Business Partner Engagement**	No Potential Investors, business partners identified. Proposal states Ohio based vendors will be used to perform work.
R	**Business Model**	Generic business model diagramed in application.
R	**Project Plan/ Budget Narrative**	Specific steps in the project have not been submitted. No proof in 1yr
Y	**Growth Plan in Ohio**	ACE is an existing Cincinnati based company; dba Wiggins Medical.
R	**ESP Interaction**	No ESP mentioned.
	Evaluator Recommendation	This application is not recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds

Comments and Recommendations: This proposal provided generic information of all the criteria. No details of specific information regarding the product, team, project plan, intellectual property, business model or any other criteria were provided. Should the company wish to reapply, it would do well to read the RFP and provide the information requested in the form of a complete and detailed proposal.

Proposal 23-4228	**Advanced & Innovative Multifunctional Materials LLC**	Amount Requested: $150,000
Licensing Institution	University of Dayton	Amount Recommended: $0
Prior Phase 1 Applications: Yes	Prior Phase 2 Applications: No	***Self-disinfecting N95 Mask***

Company Snapshot: Advanced & Innovative Multifunctional Materials LLC developed a platform technology that enables standard porous materials to possess antimicrobial properties. The first application is targeting the self-disinfecting N95 Mask market.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
Y	**Management Team**	The team consists of 3 individuals having material science, business, and technology transfer support.
R	**Opportunity/Market Size**	Market Segment is identified and quantified. The total addressable market is identified as N95 masks, however not quantified.
Y	**Intellectual Property Protection**	Provisional patent application filed in 2020, with composition of matter and method claims. PCT and WIPO filed timely.
Y	**Proof of Concept**	Phase 1 project provided a demonstration of the potential product. Data was collected from potential licensees end use customers on need.
Y	**Potential Investor/ Business Partner Engagement**	In discussion with Ohio-based equity investor focused on high-tech. AIMM is pursuing discussions with licensing targets.
Y	**Business Model**	Business model based on licensing technology to N95 mask manufacturer and collecting a royalty on each mask produced.
R	**Project Plan/ Budget Narrative**	Milestones are delineated. Funds required $101K, requesting $150K.
Y	**Growth Plan in Ohio**	AIMM has 2 FT employees, plans to grow to 14 FT & PT employees.
Y	**ESP Interaction**	AIMM is engaged with the Entrepreneurs' Center in Dayton.
	Evaluator Recommendation	This application is not recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds

Comments and Recommendations: The company has gained valuable knowledge from the Phase 1 work. The Opportunity/Market Size and Project Plan/Budget Narrative are incomplete. The company plans to license the technology and has tested the concept of attaching an extra outer layer to existing masks. Discussions with potential licensees should be clearly delineated.

Proposal 23-4233	**Phantom Solutions**	Amount Requested: $100,000
Licensing Institution	US Navy Department of Defense	Amount Recommended: $0
Prior Phase 1 Applications: No	Prior Phase 2 Applications: No	***Phantom Solutions***

Company Snapshot: Phantom Solutions is developing commercial truck fleet management software which is equipped with route optimization, and fleet/charge monitoring, and tools necessary in monetization of fleet derived carbon credit trading.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
R	Management Team	Management team is comprised of university students. Minimal work experience in relevant technologies, business/project management, start-up fund raising. Sustained commitment of management team to the project is considered uncertain.
R	Opportunity/Market Size	Proposal does not clearly identify the target markets or include Total Addressable Market (TAM) for such markets. Market size and timing of electric fleet adoption and carbon credits is uncertain.
Y	Intellectual Property Protection	There are many competitive route optimization programs with support of strong technical companies. It is not clear that appreciable competitive advantage can be achieved with carbon credit feature.
R	Proof of Concept	Proposal indicates that the team has developed "a fleet management software". However, project work plan indicates a significant amount of work remains to specify, design, and build a prototype.
R	Potential Investor/ Business Partner Engagement	Proposal does not discuss any appreciable consideration or outreach to potential investors or business partners.
Y	Business Model	Equipment sales plus SaaS for service/updates plus commission on carbon credit. GM and NM high at 92% and 52%, respectively.
R	Project Plan/ Budget Narrative	Project plan appears aggressive given current state of technology.
Y	Growth Plan in Ohio	Proposal includes projected new hires in Ohio.
Y	ESP Interaction	Discussions with The Entrepreneurs Center in Dayton, Ohio
	Evaluator Recommendation	This application is not recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds

Comments and Recommendations: Management team does not have appreciable experience or track record of business management, technical marketing and sales, S/U company formation (fund raising). Uncertain level of commitment to project and business. Proposal does not clearly describe the business opportunity with Total Addressable Market (TAM) including number of potential unit sales/subscriptions and unit prices for target markets. Proposed fleet management business opportunity is considered risky as there exist multiple software packages from well capitalized companies and also uncertain timing for adoption of electric trucks and carbon credit markets. Project timeline appears aggressive.

Proposal 23-4234	**Raider Technologies LLC**	Amount Requested: $100,000
Licensing Institution	Air Force Research Laboratory	Amount Recommended: $0
Prior Phase 1 Applications: No	Prior Phase 2 Applications: No	***Low C-SWaP Radar UAS Sensing and Monitoring System***

Company Snapshot: Raider Technologies LLC leverages a unique frequency diverse array (FDA) radar architecture that is significantly lower in cost and complexity for acquisition, detection and tracking of targets.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
R	Management Team	CEO is an accomplished program manager and technical innovator. His track record as an entrepreneur is not clear. Adding a committed entrepreneur to the Raider team / board that is focused on sales and strategic partnerships is strongly encouraged.
G	Opportunity/Market Size	Significant multi-billion-dollar opportunity in rapidly growing area. Full penetration requires new FCC regulation.
R	Intellectual Property Protection	Foundational patent expires in 2025, so will be of little help given the complexity of selling to government and need for regulatory tailwinds. Additional filings are recommended before reapplication.
Y	Proof of Concept	POC goals are aggressive, yet plausible, given prior USAF and Converge work in this area. In addition, Converge is working at attractive rates.
Y	Potential Investor/ Business Partner Engagement	Customers, go to market partners, investors all mentioned. Supplementary materials indicate meaningful engagement.
R	Business Model	Model is suspect given the short IP runway. 75% COGS / Revenue a concern. Lastly, initial sales ramp is steep given no POC yet.
Y	Project Plan/ Budget Narrative	Prior USAF work and Converge experience make this plausible.
G	Growth Plan in Ohio	Strong commitment stated with significant employment if a success.
G	ESP Interaction	Strong ESP influence apparent in interview and written supplement.
	Evaluator Recommendation	This application is not recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds

Comments and Recommendations: This proposal targets a rapidly growing market in commercial drone aviation safety. Funding is not recommended due to concerns with the team and Intellectual Property. If the team chooses to reapply, they are encouraged to add committed entrepreneurial experience to the team and file additional Intellectual Property.

Phase 1 Revised Application from Round 30

Proposal 23-2876	**The Research Institute at Nationwide Children's Hospital**	Amount Requested: $200,000
Prior Phase 1 Applications: Yes	TVSF Phase 1 Round 32 Proposal	Amount Recommended: $200,000

TOTAL BUDGET: $400,000

Institution Snapshot: The strategic vision of the program is to advance inventions/intellectual property along the commercialization pathway with the purpose of making early inventions both more attractive and less risky to potential licensing and/or business partners.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
G	**Selection Committee**	Competent 4-member committee, all with evaluation and commercialization of early-stage life science technologies. Two new members join ESP and NCH returning members.
G	**Deal Flow; Budget Strategy**	Consistent deal flow within the NCH pipeline for TDF and subsequent TVSF potential funding.
Y	**External Participation**	External providers are emphasized in the program and validation by ESP or other outside advisors strongly encouraged. Select internal resources may be used.
Y	**Track Record**	Since 2010, 52 (43 TDF/ 9 TVSF) of 125 TDF/ TVSF applications have been funded resulting in 56% licensed or pending licensing deals.
Y	**Metrics**	Of the 9 TVSF projects funded, 3 are licensed or awaiting license execution, 4 in negotiation, 1 is closed and 1 project is ongoing.
Y	**Project Management & Experience**	Each member of the 4-person committee has fund and project management experience. An interim and final report are reviewed by the licensing associate managing the inventor.
G	**Project Selection**	A detailed process for evaluating potential TVSF candidates.
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds

Comments and Recommendation: The Research Institute at Nationwide Children's Hospital has a strong track of licensing and commercializing researcher's technologies. The project selection process is well documented. Of the 9 TVSF projects funded, 3 are licensed or awaiting license execution, 3 are in negotiation, 1 is closed and 1 project is ongoing. All members of the selection committee have evaluation and commercialization experience of early-stage life science technologies.

4) Round 32 Analysis

Figure 1 shows the proposal activity and funding recommendations by technology source for Phase 2 proposals. Cleveland Clinic Foundation was the most active with three submissions followed by two submissions from University of Dayton. Two applications from the Cleveland Clinic Foundation and one from University of Dayton are recommended for funding. One application was submitted from each of Air Force Research Laboratory, University of Maine, US Navy Department of Defense, US Army, and University of Akron Research Foundation. University of Maine, US Army and University of Akron Research Foundation were each recommended for funding of these single submittals.

Figure 1. Round 32 Funding by Technology Source

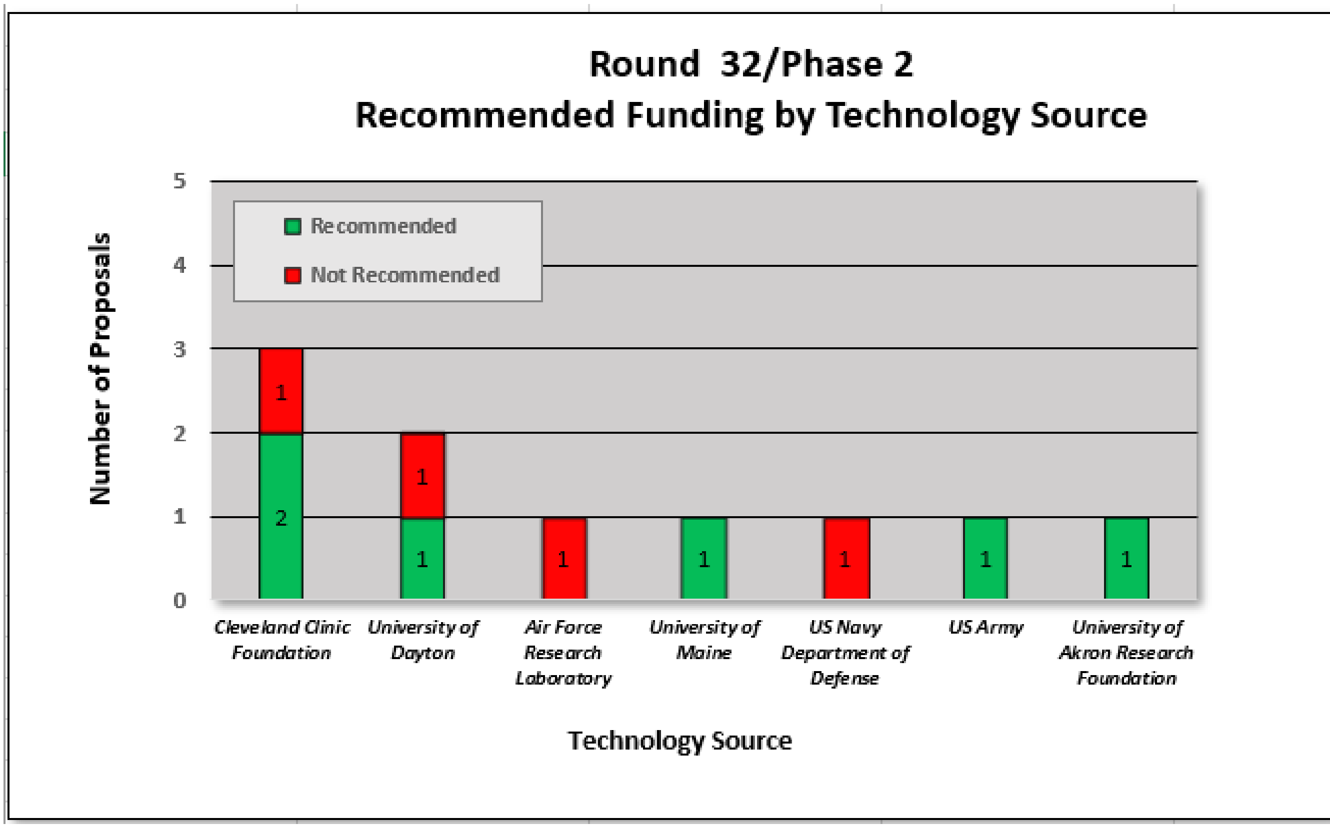


Figure 2 depicts Phase 2 proposal activity and funding recommendations by Third Frontier focus area. In this Round, five of the ten proposals were in Biomedical/Life Sciences (50%), three of ten were in Software/Information Technology (30%), one of the ten proposals were in each of Advanced Materials (10%) and Sensors (10%). Three of ten Biomedical/Life Sciences (30%) and one of the ten in each of Advanced Materials (10%), Sensors (10%), and Software/Information Technology (10%) were recommended for funding. This round represents 50% in Third Frontier Technology areas that were non-Biomedical/Life Sciences. The previous twelve rounds have averaged 56% of the applications in Biomedical/Life Sciences.

Figure 2. Round 32 Phase 2 Proposal Activity by Third Frontier Technology Area

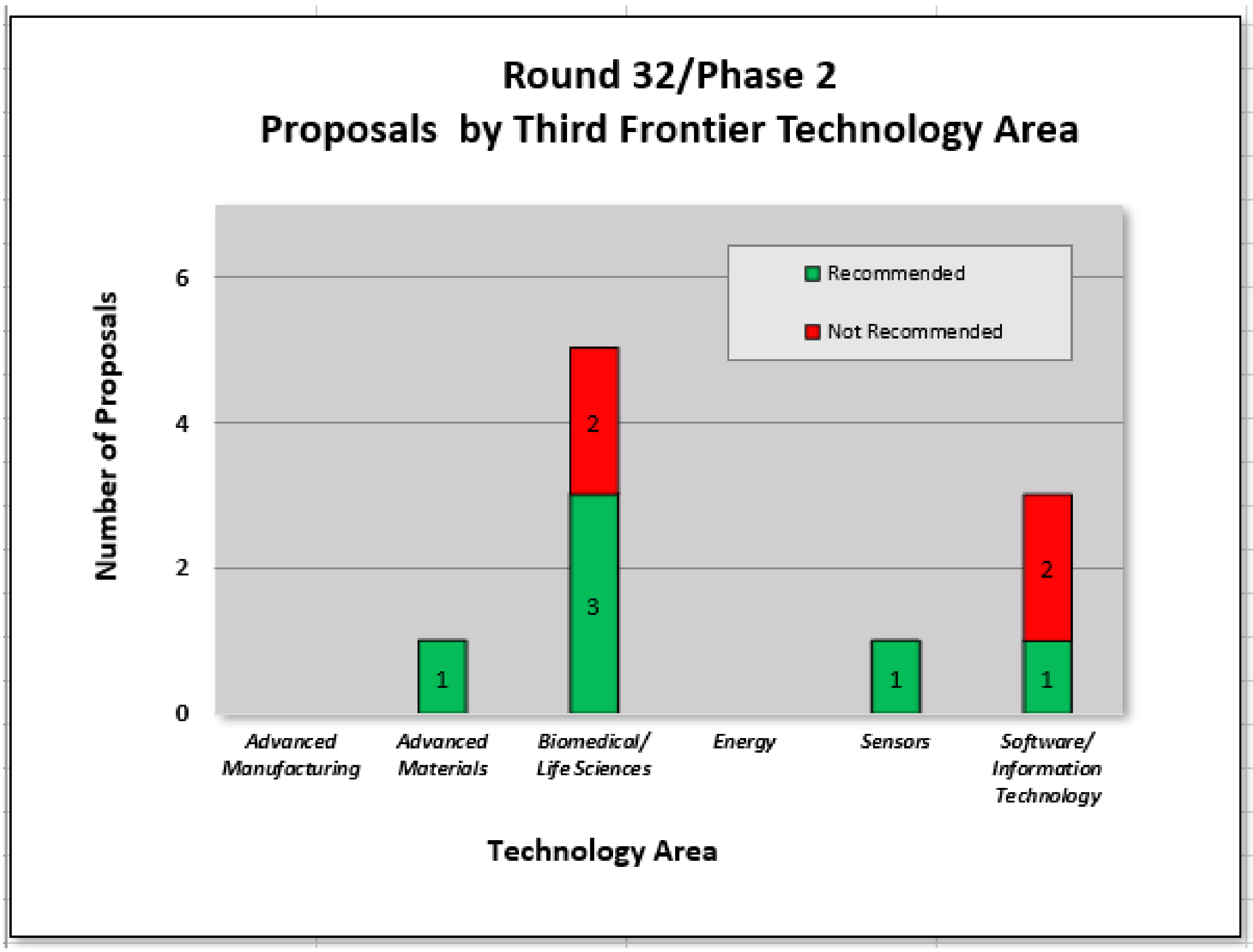
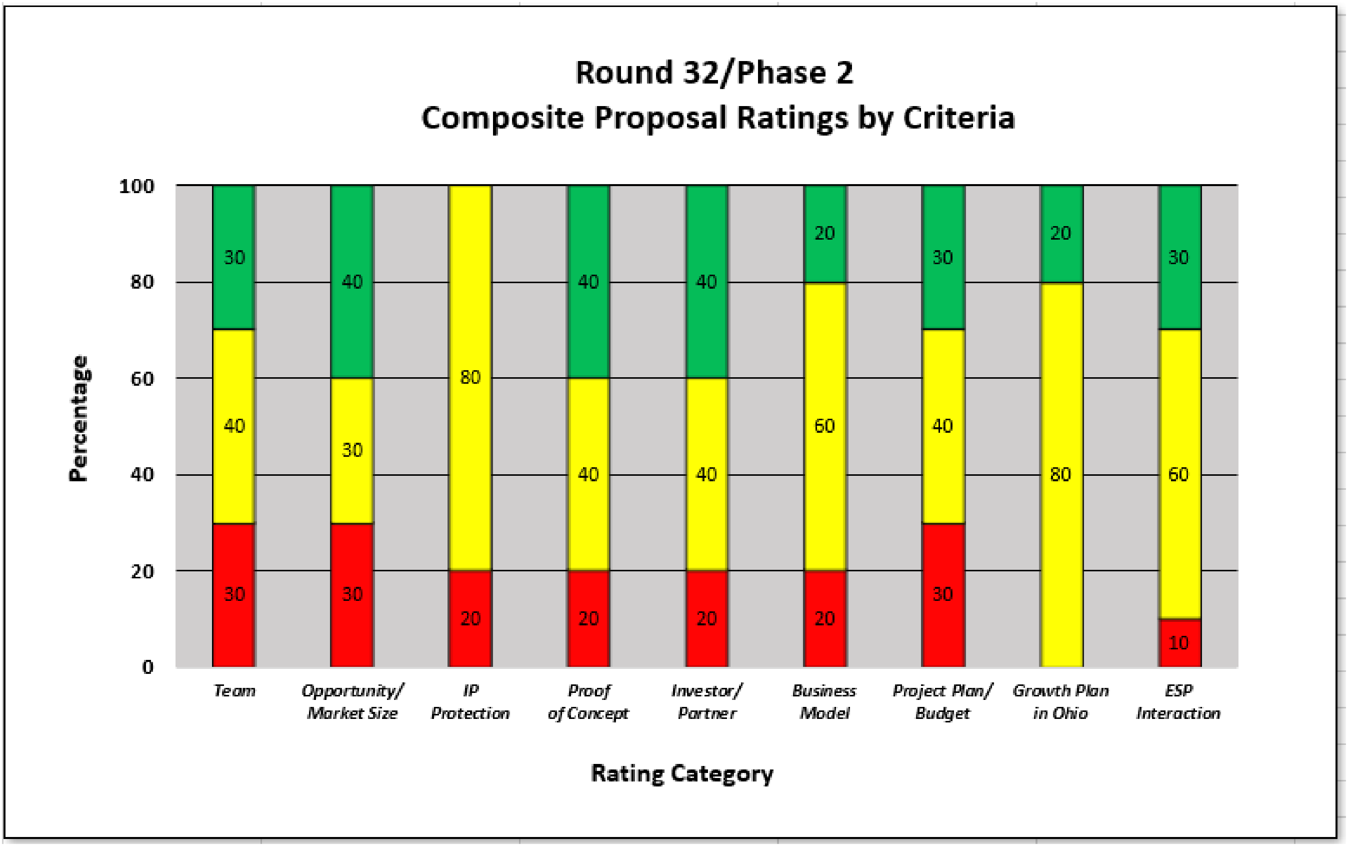


Figure 3 shows the aggregate ratings by evaluation criteria for all Phase 2 proposals. Setting aside the Growth Plan in Ohio and ESP interaction categories, the IP Protection, Proof of Concept, Potential Investor/ Business Partner Opportunity and Business Model had the lowest percentage of weak scores in this round. Team, Market Opportunity, and Project Plan/ Budget Narrative had the highest percentage of weak scores in this round.

Figure 3. Round 32 Phase 2 Proposal Rating Summary



Redwood Innovation Partners, LLC

In the previous twelve Rounds, business model was the lowest rating in Rounds 20-23 (53% average ≥ meets) and Round 26 (28%). The RFP was revised prior to Round 24 to elicit stronger business models and it appears that the proposals have provided stronger business models in Rounds 24-32 (68% average ≥ meets). Round 26 appears to be an anomaly. Further rounds will be monitored to see if the improvement continues.

Figure 4: Rounds 20 to 32 Phase 2 Analysis of Business Model

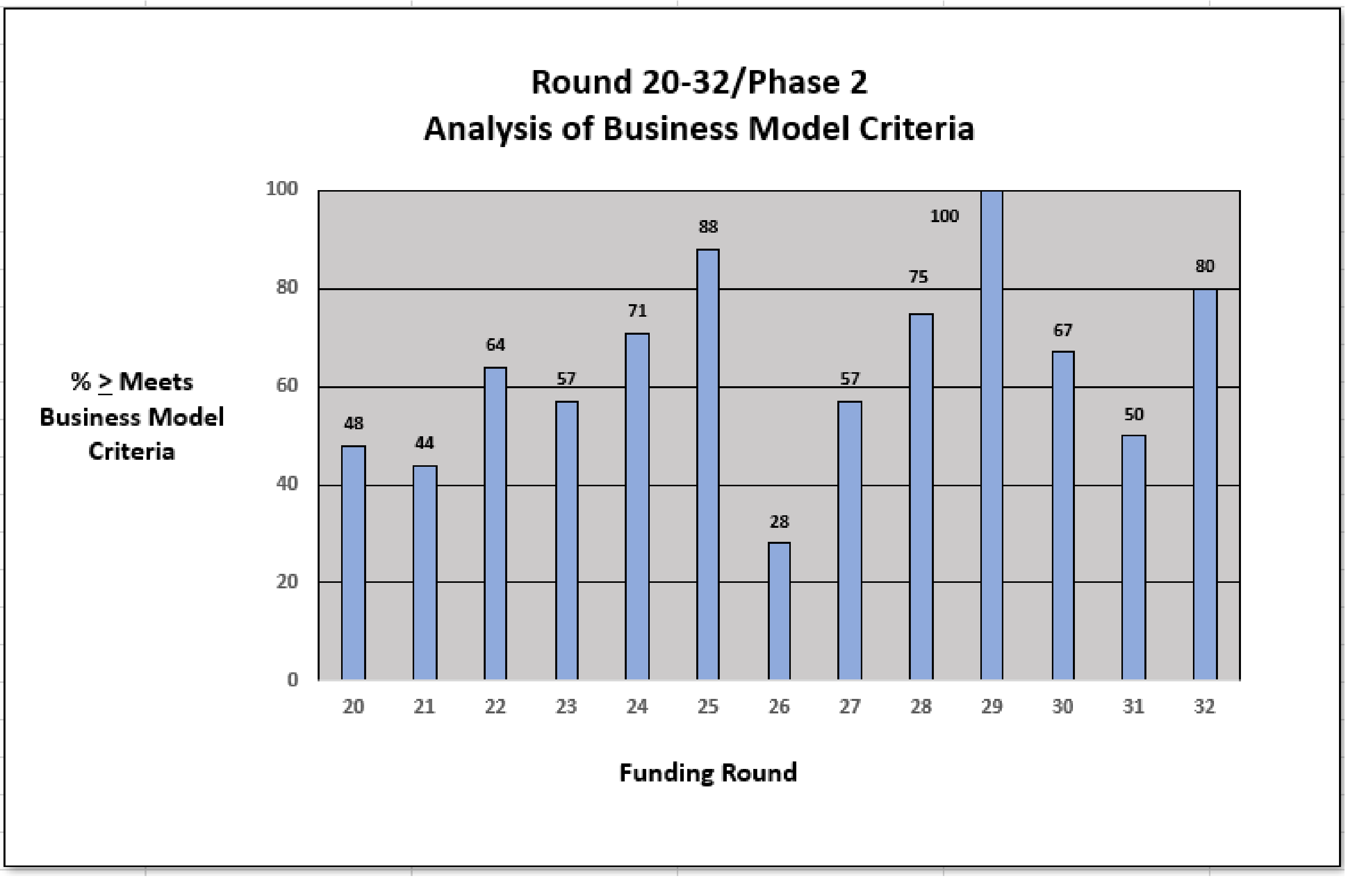
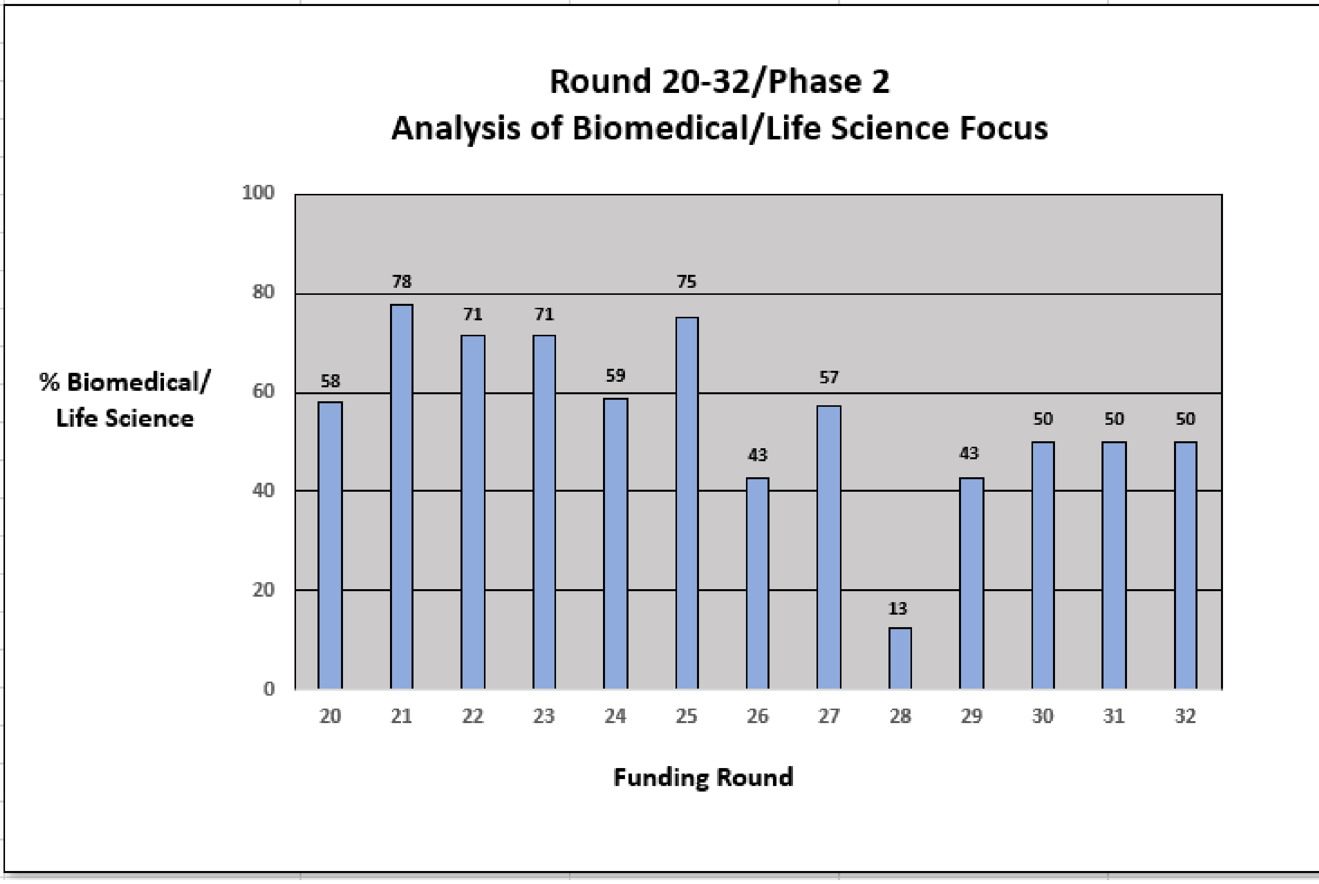


Figure 5 shows the percentage of Biomedical/Life Sciences applications for the last 13 Rounds. Round 32 represents 50% in Third Frontier Technology areas that are Biomedical/Life Sciences. Average of the last 5 rounds is 41%. All thirteen rounds average 56% of the applications in Biomedical/Life Sciences.

Figure 5: Rounds 20-32 Phase 2 Analysis of Biomedical/Life Science Focus



Carry Through and Reapplication

Phase 1 Carry Through: There was 1 Phase 2 applicant that previously received Phase 1 funding. It was not recommended for funding.

Phase 2: There were 2 reapplications in this Round. Both were first time reapplications and both were recommended for funding.

Appendix I

Summary of Redwood team and qualifications

Redwood, as a company, has been providing technology commercialization services for over 9 years while each team member has been active in this field for over 25 years.

Each Redwood team member

- possesses an advanced technical degree and extensive business proficiency
- has worked across the spectrum of technology commercialization from invention to successful market introduction
- understands how to assess a concept case from the perspective of aligning technologies to product applications in specific markets
- has lived, both conceptually and literally, the iterative process of understanding market needs and wants, value chains and who the customers are within the value chain

Team members have all worked for major corporations, research institutions, venture capital firms and technology start-up companies gaining a comprehensive understanding of what is necessary for development teams to successfully commercialize a technology. The Redwood team has served as evaluators for the Ohio Advanced Manufacturing program and an individual team member served as an evaluator for CALF, TIP and IOF loan programs for over a decade.

The five members of the Redwood team are highly qualified evaluators for the TVSF program and have combined experience and expertise in the following areas (combined years):

Commercializing technology into market pulled products (125+ years)

Market/Technology Assessment (140+ years)

Startup/ Spin out companies (50+ years)

Board member/Advisor to Startups (30+ years)

Evaluating/ monitoring RFPs/ Funding selection (40+ years)

The following is a brief summary of the five principal team members used in this evaluation Round.

Herb Bresler

- BS Biological Sciences, University of Maryland; BS Secondary Science Education, University of Maryland; PhD Immunology and Infectious Diseases, The Johns Hopkins University School of Hygiene and Public Health
- Former Senior Research Leader and Chief Scientist for Health and Life Sciences, Battelle Memorial Institute, responsible for evaluation of new technology-based business opportunities, intellectual property development, licensing and tech transfer; created and implemented new metrics to increase returns on discretionary R&D; cultivated approximately 1150 invention disclosures, 900 patent applications, and 120 granted patents, leading to $52 million company funding
- Recipient of four R&D 100 awards for breakthrough medical devices in neuroscience and diagnostics
- Former Director of the Laboratory of Cellular Immunotherapeutics at the Arthur G. James Cancer Hospital and Research Institute at The Ohio State University

John McArdle

- BE, Manhattan College, MS, Northeastern University, Chemical Engineering
- MBA, Finance / International Business, University of Chicago (Booth School of Business)
- Former Business Development Manager, Battelle
- Former Product Line Manager – Koch Industries
- Former Technical Sales Manager, Allied Signal Corporation
- Recognized expert in water and wastewater treatment technologies
- Successful track record of introducing innovative technologies for a variety of municipal, industrial, and military applications in domestic and overseas markets.

Jim Sonnett

- BS, University of Virginia, MS, University of Massachusetts, PhD, University of Delaware, all in chemical engineering
- Former Vice President – Science and Technology, Battelle Health & Life Sciences
- Former R&D Leader – W. L. Gore & Associates and E. I. DuPont
- Built and led high impact innovation organizations in aerospace, electronics, and life sciences
- Former Board Member – Velocys, Ventaira, Battelle Ventures
- Adjunct professor – Ohio State University Fisher School of Business

Susan Stanton

- BS, Millersville University, Chemistry, MPh, Syracuse University, Organic Chemistry, PhD, University of Rochester, Organic Chemistry
- Personally developed 12+ products and led new product development teams at Mobay, Alcoa & Nexicor
- Holder of 10+ patents
- Former VP Market and Technology Assessment at the National Technology Transfer Center

- Over 15 years as an angel investor in technology-based startups
- Over 15 years as an evaluator for Ohio Third Frontier funds including IOF, CALF and TIP
- Over 8 years teaching market and business analytics to STEM graduate and post doc students.

Bhima Vijayendran

- BS, University of Madras, MS, University of Madras, PhD, University of Southern California in Polymer and Surface Science, MBA, University of New Haven
- Former Senior Research Leader and Vice President Business Development, Battelle Memorial Institute; Chief Research Officer, Battelle Science and Technology, Malaysia
- Former Director, Discovery Research, PPG Industries
- Recognized as one of the leading authorities on advanced materials, special chemical and polymer systems in numerous markets including: Renewable and clean technology, Energy, Nano Technology and Industrial Products.
- Recipient of ten R&D 100 awards and over 100 patents and numerous other awards.

Appendix 2

TVSF objectives and phases

The Technology Validation and Start-up Fund (TVSF) provides grants under two phases to transition technology from Ohio Eligible Research Institutions into the marketplace through Ohio start-up companies. Under Phase 1, Ohio Research Institutions may apply for a pool of funds to support validation/ proof that will directly impact and enhance both the commercial viability of their unlicensed technologies and ability to support a start-up company. Under Phase 2, Ohio start-up and young companies may apply for funding to commercialize a technology they intend to license from a university or an Ohio research institution.

The goals of Phase 1 include:

- Generate the proof needed to move technologies to the point that they are either ready to be licensed by an Ohio start-up company or deemed unfeasible for commercialization. The institutions are encouraged to work with potential Ohio licensees to identify the proof needed.
- Perform validation activities such as demonstration and assessment of critical failure points in subsequent development, prototyping, scale-up and commercialization in order to generate this proof with strong preference for these activities being performed by an independent 3rd party source.

The goals of Phase 2 include:

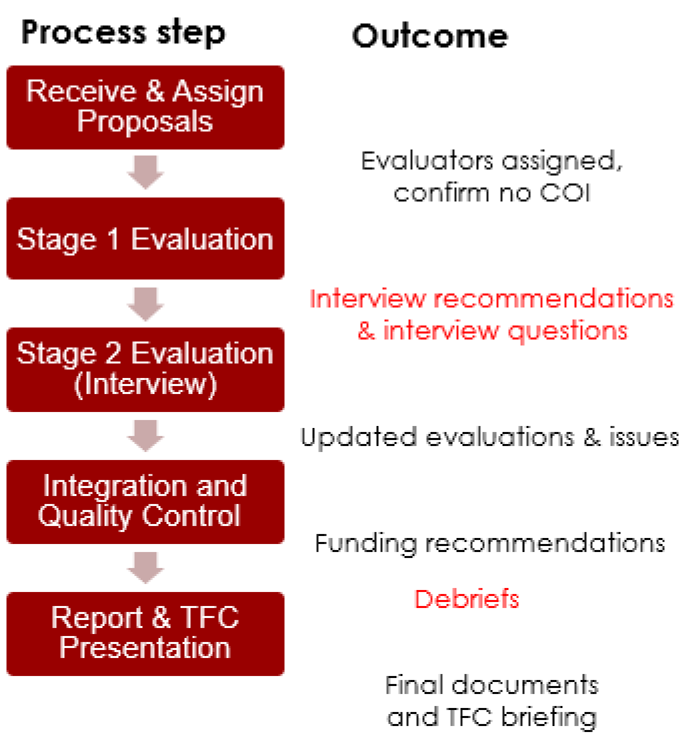
- Accelerate the commercialization of technology by Ohio start-up companies that license technology developed at Eligible Institutions during the critical early stage of life of the company.
- Generate the proof needed to move technology to the point where it is able to be commercialized or additional funds for commercialization can be raised. A clearly identified path to subsequent funding opportunities and working directly with potential investors to define the proof needed for investment into the company is strongly encouraged.
- Funded activities may include, but may not be limited to, beta prototype development and deployment to potential customers for testing and evaluation and market research/ business development in order to generate the proof needed.

Based upon these goals, the proposal evaluation criteria were developed. The proposals were then evaluated based on the criteria.

Description of review process

Review summary. Our overall review process flow and outcomes by stage are shown in Figure 1. A similar process has been successfully used by Redwood in prior projects for public and private clients. Discussions were held with the TVSF program manager after all but the initial step in Figure 1.

Figure 1. TVSF Evaluation Process



Review and Assign Proposal In this first step proposals were summarized and a primary evaluator was assigned who has the appropriate background and no conflict of interest.

Stage 1 Evaluation Stage 1 evaluations were conducted for each proposal using the criteria shown below in Tables 1 and 2. Differentially weighted criteria were used to evaluate Phase 1 and Phase 2 proposals. Each proposal was rated on a 0 (absent) – 5 (Outstanding) scale for each criterion, an approach used by the NSF and in other State of Ohio programs. The weightings reflect the experience of the Redwood team and our belief that some factors, for example team and market opportunity in Phase 2, are more important than others.

The entire review team subsequently discussed all the evaluations to ensure consistency and agreed upon which applicants to invite for interviews. Interview questions were then provided in advance to each applicant.

Stage 2 Evaluations (Interviews) The standard procedure for this step is: In-person, 45-minute interviews were held with each invited applicant to discuss the advance questions plus other topics of interest to the evaluators. Two Redwood team members participated in the interviews in person with additional team members joining via Zoom or a conference call.

Integration and Quality Control Proposal evaluations were updated based on interview results. A calibration review was held by the review team to ensure that evaluations were performed consistently and that any changes made were a result of team consensus. Based on this review, proposals were recommended for funding.

Table 1 – Phase 1 Evaluation Criteria

Criterion	Weighting	Description
Alignment and Compliance	Go / No go	Institutional alignment with TVSF intent and compliance with RFP
Project Selection Committee	20	Skills, background and commitment of the committee members
Deal Flow; Budget Strategy	15	Is the projected deal flow consistent with the requested budget to enable committing funds within 1 year?
External Participation	15	Does process ensure validation activities will be performed by 3rd parties; ESPs and state-funded programs/organizations are enlisted to enhance commercialization activities of the project?
Track Record	15	Is there a strong Phase 1 or comparable program track record of licensing and newco creation? If not, is there a plan for improvement?
Metrics	15	Realism and impact of proposed metrics, including licensing, start-ups.
Project Management & Experience	15	Is there a strong project management strategy and appropriate experience of people who allocate the pool of funds and manage individual projects?
Project Selection Process	5	Is there a clear, appropriate process for project selection?

Table 2 – Phase 2 Evaluation Criteria

Criterion	Weighting	Description
Alignment & compliance	Go / No Go	Proposal alignment with TVSF intent and compliance with RFP
Management Team	20	Skills, background and commitment
Opportunity / market size	15	What is the market segment and total addressable market? Is it a platform or breakthrough technology or incremental improvement? If breakthrough, is it compatible with viable commercialization pathways?
IP Protection / License	15	Is IP adequately protected, does it enable the business model, is it differentiated from likely competition, is license likely within 9 months?
Compelling Proof of Concept	15	Was meaningful input from potential customers and key performance metrics used to design Proof of Concept? Are the competitive advantages compelling for potential customers?
Potential Investor / Business Partner Engagement	10	Is there company engagement / collaboration independent of licensing institution, including financial backing?
Business Model	10	Is the business model realistic AND achievable? Can the service / manufacturing model be scaled?
Project Plan / Budget Narrative	5	Is the budget consistent with proof in 1 year?
Start-up in Ohio	5	Does a start-up exist or is it planned? Will the start-up be in Ohio?
ESP Interaction	5	Is team engaged with ESP? Has team incorporated feedback from ESP into the project, proposal or business plan?